

Review – 1:

Title: Explainable AI Techniques for Student Academic Performance Analytics: A Comparative Study

1. Problem Statement

Machine learning models for predicting student academic performance have achieved high predictive accuracy but often function as "black boxes," providing little insight into the factors driving their predictions. This lack of interpretability limits trust among educators and administrators, hindering the adoption of data-driven decision-making in educational settings . While Explainable AI (XAI) techniques such as SHAP and LIME have emerged to address this gap, there remains a need for comprehensive comparative evaluations of these techniques across multiple machine learning models and educational datasets to determine which approaches provide the most actionable and trustworthy explanations for diverse educational stakeholders.

2. Literature Survey

S.No	Title / Authors / Year	Source	Key Contributions
1	"Model Interpretability in the Educational Data Mining Context: A Systematic Literature Mapping" – Carvalho, Mattos, & Aguiar (2026)	Revista Brasileira de Informática na Educação	Found that interpretability is frequently addressed as a post-hoc component rather than as a core objective; identified need for comparative studies across different ML techniques

2	"Systematic Literature Review on Explainable Learning Analytics and Educational Data Mining" – IEEE Access (2025)	IEEE Access (Vol. 13)	Reviewed 212 studies; revealed researchers have increasingly adopted post-hoc methods—especially SHAP—from 2024 to 2025; identified gaps in quantifiable explainability metrics
3	"Leveraging psychological data and transparent machine learning for student success prediction" – ScienceDirect (2026)	ScienceDirect	Evaluated six ML classifiers; XAI highlighted digital self-efficacy, perceived feedback usefulness, and discussion engagement as dominant predictors
4	"Ensemble machine learning framework with SHAP and LIME for accurate early prediction of student success" – Almazroei (2026)	Scientific Reports / NIH	Integrated 7 datasets; XGBoost achieved 95.04% accuracy; SHAP/LIME revealed cumulative assessment performance and engagement as top predictors
5	"Predicting student success with interpretable machine learning framework" – Frontiers in Education (2026)	Frontiers	CatBoost achieved 88.8% balanced accuracy; XAI enabled global and individual-level interpretation; emphasized socioeconomic factors
6	"Explainable student performance prediction models: A systematic review" – Alamri & Alharbi (2021)	IEEE Access	Comprehensive review of XAI in education; highlighted SHAP and LIME as dominant techniques; identified lack of fairness and interpretability metrics
7	"Tracking student progress using clustering and explainable AI" – Springer (2026)	Universal Access in the Information Society	Proposed rule-based system transforming XAI rules into human-readable feedback; LLM-assisted recommendation generation

8	"Explainable AI in education: A survey" – Khosravi et al. (2022)	Computers and Education: AI	Provided foundational framework for XAI in educational contexts; discussed stakeholder-specific explanation requirements
9	"Analyzing dropout and explainable prediction using AI" – Paul et al. (2025)	Frontiers in Education	Logistic Regression found most effective for at-risk identification; XAI provided explainable insights into decision-making
10	"Predicting academic performance through lifestyle factors using interpretable ML" – World Scientific (2026)	International Journal on AI Tools	Proposed hierarchical evolutionary feature selection framework balancing predictive accuracy with interpretability
11	"A Novel Hybrid Feature Selection Framework for Enhancing Accuracy and Interpretability" – Bin Azmi et al. (2025)	International Journal of Applied Mathematics	Proposed LASSO + ACO hybrid feature selection to balance accuracy and interpretability; highlighted "black-box" decision transparency
12	"From Predictive Analytics to Explainable AI in Higher Education" – Talmoudi & Choukir (2026)	Qubahan Academic Journal	Framework integrating predictive analytics and XAI for higher education; emphasis on transparent decision-making

4. Research Gaps Identification

Based on the literature review, the following research gaps have been identified:

S.No	Research Gap	Source
1	Interpretability as Post-Hoc Component – Most studies treat interpretability as an afterthought rather than a core objective of model design; limited systematic evaluation and comparative analysis of interpretability across models from different ML techniques	Carvalho et al. (2026)
2	Limited Quantifiable Explainability Metrics – Only a few studies include quantified explanations using metrics like sensitivity and stability; lack of standardized evaluation frameworks for XAI in education	IEEE Access SLR (2025)
3	Lack of Comparative Studies – Need for studies comparing interpretability across multiple ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost) using identical datasets and XAI techniques	Carvalho et al. (2026)

4	Limited Fairness Integration – Fairness and equity considerations remain underexamined in XAI educational research; need to assess group-level fairness in predictive systems	ScienceDirect (2026), Paul et al. (2025)
5	Limited Multi-Dataset Integration – Most studies rely on single datasets; need for comprehensive comparison across multiple benchmark datasets (UCI, xAPI, OULAD)	Almazroei (2026)

5. Objectives

S.No	Objective	Technique to Achieve
1	To evaluate and compare the predictive accuracy of five machine learning models (Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost) for student academic performance classification	Use Scikit-learn, XGBoost, CatBoost with 5-fold cross-validation and SMOTE for class imbalance
2	To apply and compare SHAP and LIME XAI techniques for providing global and local explanations across all models	Use SHAP (TreeExplainer, KernelExplainer) for global explanations and LIME for local explanations; compare explanation quality and consistency

3	To identify the most influential academic, behavioral, and demographic factors affecting student performance	Use SHAP summary plots, force plots, and dependence plots to rank feature importance across all models
4	To conduct a comparative analysis of models based on performance metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) and interpretability scores	Use Scikit-learn metrics and Borda Count method for model selection; create Streamlit/Gradio dashboard for interactive visualization

6. Novelty / Innovation

S.No	Innovation	Description
1	Multi-Dataset Integration Pipeline	Integrating three benchmark datasets (UCI, xAPI, OULAD) into a unified preprocessing and analysis pipeline—rarely done in previous studies
2	Comprehensive XAI Comparison Framework	Systematic side-by-side comparison of SHAP and LIME across 5 models using identical feature sets and evaluation criteria—addressing the gap identified by Carvalho et al. (2026)

3	Feature-Level Interaction Analysis	Using SHAP dependence plots to reveal how features interact (e.g., attendance × GPA) in influencing predictions—beyond what previous studies have explored
4	Explainability-Quantified Dashboard	Developing an interactive dashboard where explanations are paired with confidence metrics—addressing the gap in quantifiable explainability metrics identified by the IEEE SLR (2025)

7. Feasibility

Aspect	Assessment
Data	All three datasets (UCI, xAPI, OULAD) are publicly available and well-documented; OULAD available via UCI Repository
Technical	All tools (Python, Scikit-learn, XGBoost, CatBoost, SHAP, LIME, Streamlit) are open-source and well-supported
Computational	Models are computationally feasible on standard hardware; SHAP TreeExplainer optimized for tree-based models

8. Project Plan

Phase 1: Data Collection & Preprocessing

- Download and merge UCI, xAPI, OULAD datasets
- Handle missing values, encode categorical variables, normalize features
- Apply SMOTE for class imbalance

Phase 2: Exploratory Data Analysis (EDA)

- Perform 12+ EDA techniques: dataset overview, descriptive statistics, missing value analysis, duplicate detection, outlier detection, univariate/bivariate/multivariate analysis, correlation analysis

Phase 3: Feature Engineering & Selection

- Create new features from existing data
- Select optimal features using correlation and importance ranking

Phase 4: Model Training

- Train 5 models: Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost
- Use 5-fold cross-validation and hyperparameter tuning

Phase 5: Performance Evaluation

- Evaluate using Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Apply Borda Count method for model ranking

Phase 6: XAI Implementation

- Apply SHAP (global summary plots, force plots, dependence plots) across all models
- Apply LIME for local explanations on individual predictions

Phase 7: Comparative Analysis & Documentation

- Compare models on accuracy vs. interpretability trade-off
- Compare SHAP vs. LIME for explainability quality
- Document findings in comprehensive project report
- Record project demo video